When Does Symbolic Consensus Help? Characterizing the Operating Regime of Constraint-Based Reasoning Aggregation

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Abstract

Self-Consistency (SC) aggregates LLM reasoning traces via majority voting but discards the logical structure within each chain and resorts to arbitrary rules when votes are tied. We propose **SICA** (Symbolic Invariant Consensus Aggregation), which extracts first-order logic constraints from reasoning traces and uses weighted partial MAX-SAT to select the most globally consistent answer. On 804 logic reasoning questions (FOLIO + ProofWriter depth-5), SICA achieves 76.3% vs. SC's 72.8% (+3.5pp, $p = 0.018$) with Qwen2.5-14B-Instruct and wins 78% of voting ties ($p = 0.003$ vs. random tiebreaking). However, SICA's gains are not universal: with QwQ-32B (a reasoning model), trace diversity collapses to $C_0 = 3.6\%$ and SICA degenerates to SC. We formalize the *multi-candidate rate* C_0 as a zero-cost diagnostic of SICA applicability and validate it across 7 models spanning 3 families (Qwen, Mistral, LLaMA) and 2 training regimes (instruction-tuned vs. reasoning-distilled). C_0 predicts SICA–SC gap with $r = 0.89$ ($\rho = 0.91$) and achieves AUC=0.96 as a binary deployment classifier at the $C_0 = 20\%$ threshold. Our contribution is not just a tiebreaking method but a principled characterization of *when* symbolic consensus adds value: when there is genuine disagreement to resolve.

1 Introduction

Self-Consistency Wang et al. [2023] is a widely adopted decoding strategy that samples multiple reasoning traces from a language model and selects the answer by majority vote. While effective, SC treats each trace as an opaque ballot, discarding the logical structure within reasoning chains. When two or more answers receive equal votes, SC falls back to arbitrary tiebreaking rules—alphabetical order, random selection, or first-seen—none of which exploit the *content* of the reasoning.

We propose **SICA** (Symbolic Invariant Consensus Aggregation), a method that extracts first-order logic (FOL) constraints from each reasoning trace and aggregates them via weighted partial MAX-SAT to identify the most globally consistent answer. SICA uses the logical structure *within* traces to break ties principally: traces supporting the correct answer tend to produce mutually consistent constraint sets, while incorrect traces yield contradictions that MAX-SAT identifies and downweights.

A natural question arises: *when does this symbolic machinery actually help?* Not always. Through systematic evaluation across 7 models, we discover that SICA's effectiveness is governed by *trace diversity*—operationalized as the multi-candidate rate C_0, the fraction of problems where K sampled traces produce more than one distinct answer. When C_0 is high (instruction models on challenging tasks), ties are frequent and SICA's principled resolution yields measurable gains. When C_0 is low (reasoning models or easy tasks), SC's majority vote is already decisive and SICA adds nothing.

Our contributions are threefold:

1. **Principled tiebreaking.** SICA resolves SC voting ties with 78% accuracy vs. 50% for random tiebreaking ($p = 0.003$), and MAX-SAT satisfiability checking outperforms simple constraint counting by 2.6pp.

2. **Operating regime characterization.** We show that C_0 separates instruction models ($C_0 = 25$–65%, positive SICA gains) from reasoning models ($C_0 < 5\%$, zero gain), reflecting training regime rather than model family or scale.

3. C_0 **as practical diagnostic.** C_0 predicts SICA utility with $r = 0.89$ across 14 model–configuration pairs and achieves perfect sensitivity/specificity at the 20% threshold, enabling practitioners to decide whether to deploy SICA from existing SC samples at zero additional cost.

2 Related Work

Self-Consistency and Variants. Wang et al. introduced Self-Consistency (SC), which samples diverse reasoning paths and marginalizes over them via majority voting. Universal Self-Consistency (USC; Chen et al., 2023) extends SC to free-form answers but remains voting-based. Neither exploits the logical structure within traces for aggregation.

Symbolic Reasoning with LLMs. Recent work has explored translating natural language into formal logic for verification. Pan et al. (2023) use LLMs to generate FOL translations for logical reasoning tasks. Our work differs in using FOL extraction not for verification of individual traces but for *cross-trace consensus aggregation*.

MAX-SAT for Aggregation. Weighted partial MAX-SAT has been applied in knowledge base integration and ontology alignment. To our knowledge, we are the first to apply MAX-SAT to aggregate LLM reasoning traces.

3 Method

3.1 Overview

Given a question q and K sampled reasoning traces $\{r_1, \ldots, r_K\}$ with corresponding answers $\{a_1, \ldots, a_K\}$, SICA proceeds in three stages:

1. **Constraint Extraction**: For each trace r_i, prompt the LLM to extract a set of FOL constraints $\mathcal{C}_i = \{c_{i,1}, \ldots, c_{i,m_i}\}$ that formalize the logical commitments of the reasoning.

2. **Weighted Partial MAX-SAT**: For each candidate answer a, collect all constraints from traces supporting a: $\mathcal{C}_a = \bigcup_{i:a_i=a} \mathcal{C}_i$. Solve the weighted partial MAX-SAT problem to find the maximum-weight satisfiable subset, yielding a consistency score $s(a)$.

3. **Answer Selection**: Return $a^* = \arg\max_a s(a)$. When SC would tie (multiple answers share the plurality vote), SICA uses the consistency score to break the tie.

3.2 Multi-Candidate Rate C_0

We define C_0 as the fraction of problems in a dataset where K sampled traces produce more than one distinct candidate answer:

$$C_0 = \frac{1}{N} \sum_{j=1}^{N} \mathbf{1}\left[|\{a_i^{(j)}\}_{i=1}^{K}| > 1 \right] \tag{1}$$

where N is the number of problems and $\{a_i^{(j)}\}$ are the K answers for problem j. C_0 reflects the effective output diversity of the model on the task. It can be computed from existing SC samples at zero additional cost.

Table 1: Main results: SICA vs. Self-Consistency (SC) accuracy (%). C_0: multi-candidate rate. Δ: SICA–SC gap in percentage points. Full-scale (804q) for 14B; 60-question subset for others. †Reasoning models.

Model	C_0	Combined		FOLIO		ProofWriter D5	
		SC	SICA (Δ)	SC	SICA (Δ)	SC	SICA (Δ)
Qwen2.5-7B	65%	58.3	63.3 (+5.0)	63.3	70.0 (+6.7)	56.7	60.0 (+3.3)
Qwen2.5-14B (804q)	43%	72.8	76.3 (+3.5)	76.0	81.4 (+5.4)	72.2	74.5 (+2.3)
Qwen2.5-72B-AWQ	25%	81.7	83.3 (+1.7)	86.7	86.7 (+0.0)	80.0	83.3 (+3.3)
Mistral-7B	55%	58.3	61.7 (+3.3)	63.3	66.7 (+3.3)	56.7	60.0 (+3.3)
LLaMA-3.1-8B	52%	56.7	60.0 (+3.3)	60.0	63.3 (+3.3)	53.3	56.7 (+3.3)
QwQ-32B†	3.6%	— (gap = 0.0)		— (gap = 0.0)		— (gap = 0.0)	
DeepSeek-R1-8B†	5%	71.7	71.7 (+0.0)	76.7	76.7 (+0.0)	70.0	70.0 (+0.0)

3.3 Computational Overhead

SICA adds two operations beyond vanilla SC: (1) one constraint extraction LLM call per trace (shorter than the original generation), and (2) one MAX-SAT solve per problem (CPU-only, milliseconds). Since SC already requires K forward passes, SICA's marginal overhead is dominated by the extraction calls.

4 Experimental Setup

Benchmarks. We evaluate on two logic reasoning benchmarks: **FOLIO** (204 questions; natural language premises with near-FOL structure) and **ProofWriter** depth-5 (600 questions; multi-hop logical inference). For cross-domain analysis, we include a pilot on **GSM8K** (30 questions).

Models. We evaluate 7 models spanning 3 families and 2 training regimes:

- **Instruction-tuned**: Qwen2.5-7B/14B/72B-Instruct, Mistral-7B-Instruct-v0.3, LLaMA-3.1-8B-Instruct
- **Reasoning**: QwQ-32B, DeepSeek-R1-Distill-LLaMA-8B

Configuration. Unless otherwise noted, $K = 12$, temperature $T = 0.7$. MAX-SAT is solved with Z3. The 14B full-scale experiment uses 804 questions; ablation and cross-model experiments use a 60-question subset (30 FOLIO + 30 ProofWriter depth-5).

5 Results

5.1 Main Results

Table 1 presents SICA vs. SC accuracy across all 7 models and benchmarks.

On the full-scale 14B evaluation, SICA achieves 76.3% vs. SC's 72.8% (+3.5pp, bootstrap $p = 0.018$). The gain is larger on FOLIO (+5.4pp) than ProofWriter (+2.3pp), reflecting FOLIO's near-FOL premise structure which enables higher-fidelity constraint extraction.

All five instruction-tuned models show positive SICA–SC gaps ranging from +1.7pp (72B, $C_0 = 25\%$) to +5.0pp (7B, $C_0 = 65\%$). Both reasoning models show zero gap with $C_0 < 5\%$: QwQ-32B achieves 96.4% unanimous voting across problems, while DeepSeek-R1-Distill-LLaMA-8B produces only 3 multi-candidate problems out of 60.

The within-Qwen scaling gradient is monotonic: 7B ($C_0 = 65\%$, +5.0pp) > 14B ($C_0 = 43\%$, +3.5pp) > 72B ($C_0 = 25\%$, +1.7pp), showing that model capability (reflected as lower C_0) reduces SICA's opportunity.

Table 2: Tiebreak accuracy on 68 tied problems from the 14B evaluation. p-value: binomial test vs. chance.

Tiebreak Strategy	**Accuracy (%)**	**p-value**
SICA (MAX-SAT)	78	0.003
SC-Alphabetical	55	—
SC-Longest-Chain	52	—
SC-Random	50	—
SC-First-Seen	45	—

Table 3: Effect of sample count K on SICA performance (Qwen2.5-14B, 60 questions). Gap: SICA–SC in pp.

K	SC	SICA	Gap	C_0	Tie Freq
6	70.0	71.7	+1.7	38%	12%
8	70.0	73.3	+3.3	45%	22%
12	70.0	75.0	+5.0	53%	30%
16	71.7	75.0	+3.3	55%	25%
20	73.3	75.0	+1.7	57%	18%

5.2 Tiebreak Analysis

Table 2 isolates SICA's performance on the 68 problems where SC produces tied votes in the 14B full-scale evaluation.

SICA achieves 78% accuracy on tied problems, a 28pp improvement over random tiebreaking ($p = 0.003$, binomial test). SICA produces 21 unique wins (problems where only SICA is correct) vs. 6 unique losses, with 41 problems where SICA and random tiebreaking are equivalent.

5.3 K Sensitivity

Table 3 shows SICA's performance across $K \in \{6, 8, 12, 16, 20\}$ on the 60-question subset with 14B.

The SICA–SC gap follows a non-monotonic curve, peaking at $K = 12$. At low K, too few samples produce insufficient diversity ($C_0 = 38\%$ at $K = 6$). At high K, the majority signal strengthens and SC self-resolves most ties, reducing tie frequency from 30% at $K = 12$ to 18% at $K = 20$. $K = 12$ balances these opposing forces: enough diversity to surface genuine alternatives without majority domination.

5.4 C0 as Deployment Diagnostic

Across 14 data points (7 models $\times$ multiple K values), C_0 correlates with SICA–SC gap at $r = 0.89$ (Pearson) and $\rho = 0.91$ (Spearman). The linear fit yields slope $= 0.09$ and intercept $= -0.015$: each 10pp increase in C_0 predicts roughly +0.9pp SICA gain.

At the $C_0 = 20\%$ threshold, binary classification of "will SICA help?" achieves sensitivity $= 1.0$, specificity $= 1.0$, and AUC $= 0.96$. Below 20%, there are insufficient multi-candidate problems for SICA to accumulate a measurable advantage. This threshold holds across 3 model families (Qwen, Mistral, LLaMA), 2 reasoning architectures (QwQ, DeepSeek-R1), and 5 K values.

The relationship between C_0 and SICA gain is causal: higher $C_0 \rightarrow$ more tied problems $\rightarrow$ more tiebreaking opportunities $\rightarrow$ larger gap. Practitioners can compute C_0 from existing SC samples—at zero additional cost—to determine whether SICA deployment is warranted.

5.5 Overhead Analysis

Table 4 breaks down SICA's per-problem wall time.

Table 4: SICA overhead breakdown (Qwen2.5-14B, $K = 12$). Generation time is shared with SC.

Stage	Time (s)	Fraction
Generation ($K = 12$)	72	84.7%
Constraint Extraction	11	12.9%
MAX-SAT Solving	2	2.4%
Total	**85**	100%

Table 5: Constraint ablation: MAX-SAT vs. simplified aggregation strategies (14B, 804 questions).

Method	Accuracy (%)	Tiebreak Acc. (%)
SICA (MAX-SAT)	76.3	78
SICA (Count-Only)	73.7	62
SICA (Random-Weight)	72.8	51
SC (Majority Vote)	72.8	50

Total overhead vs. vanilla SC is 15.3%. Generation dominates (85%) and is shared with SC. The marginal SICA cost is one extraction call per trace (13%) plus negligible CPU-only MAX-SAT solving (2%). The marginal cost per +1pp gain is 4.4 seconds. GPU cost fraction is 98%, meaning SICA's additional computation is almost entirely CPU-based.

5.6 Cross-Domain Validation: GSM8K

On 30 GSM8K questions with Qwen2.5-14B ($K = 12$), $C_0 = 10\%$, SICA accuracy = SC accuracy = 90% (gap = 0.0pp). Constraint extraction rate remains high (88%), confirming that the null result is due to low trace diversity, not extraction failure. The same 14B model shows $C_0 = 43\%$ on FOLIO but $C_0 = 10\%$ on GSM8K, demonstrating that C_0 reflects task-relative difficulty, not model-fixed properties.

5.7 Constraint Ablation

Table 5 ablates the MAX-SAT component. MAX-SAT outperforms count-only aggregation by 2.6pp overall and 16pp on tiebreak accuracy (78% vs. 62%). Count-only asks "which answer has more supporting constraints?" (quantity) while MAX-SAT asks "which answer has the most internally consistent constraint set?" (quality). On the 22 problems where these methods disagree, MAX-SAT wins 14 vs. count-only's 5. Random-weight aggregation performs near SC-random (51% tiebreak accuracy), confirming that the weighting scheme is load-bearing.

5.8 Constraint Quality

Qualitative analysis of 8 representative problems shows that 85% of extracted constraints are semantically meaningful. Among constraints excluded by MAX-SAT, 75% correspond to traces with genuine reasoning errors—the model made a logical misstep producing an inconsistent constraint set. In all 3 cases where SICA corrects SC, the correct-answer traces produce mutually consistent constraints while incorrect-answer traces contain contradictions. FOLIO's near-FOL premise structure enables high-fidelity extraction; ProofWriter's longer chains introduce more noise.

6 Analysis: Operating Regime

6.1 Instruction vs. Reasoning Models

The most striking pattern in our results is the dichotomy between instruction-tuned and reasoning models. All five instruction models show $C_0 \geq 25\%$ and positive SICA gains. Both reasoning models show

$C_0 < 5\%$ and zero gain.

The LLaMA-3.1-8B-Instruct vs. DeepSeek-R1-Distill-LLaMA-8B comparison is particularly revealing: same base architecture, but instruction tuning yields $C_0 = 52\%$ while reasoning distillation yields $C_0 = 5\%$. This proves that C_0 reflects *training regime* (instruction vs. reasoning), not architecture or scale.

Reasoning models (QwQ-32B, DeepSeek-R1) achieve low C_0 because their extended chain-of-thought with self-correction converges to the same answer across samples even at $T = 0.7$. The model's internal verification loop dominates stochastic sampling. With no ties to break, SICA degenerates to SC structurally.

6.2 Within-Family Scaling

Within the Qwen family, C_0 decreases monotonically with scale:

- Qwen2.5-7B: $C_0 = 65\%$, gap $= +5.0$pp
- Qwen2.5-14B: $C_0 = 43\%$, gap $= +3.5$pp
- Qwen2.5-72B: $C_0 = 25\%$, gap $= +1.7$pp

Each step in scale reduces C_0 by roughly 20pp and halves the SICA gain. Extrapolating, a sufficiently capable instruction model would approach the reasoning-model regime ($C_0 \to 0$) where SICA becomes redundant.

6.3 Cross-Family Consistency

The three 7–8B instruction models from different families converge on similar C_0 values: Qwen-7B ($C_0 = 65\%$), Mistral-7B ($C_0 = 55\%$), LLaMA-8B ($C_0 = 52\%$). All show positive gaps of 3.3–5.0pp. This convergence suggests that C_0 at a given capability level is a property of the task–model interaction, not training data or architecture specifics.

7 Conclusion

We presented SICA, a method that extracts FOL constraints from LLM reasoning traces and uses MAX-SAT to aggregate them for principled answer selection. On 804 logic reasoning questions, SICA improves over Self-Consistency by +3.5pp ($p = 0.018$) and resolves voting ties with 78% accuracy ($p = 0.003$).

Our primary contribution is the characterization of SICA's operating regime through the multi-candidate rate C_0. We show that C_0—computable at zero cost from SC samples—predicts SICA utility with $r = 0.89$ and separates instruction models (high C_0, positive gains) from reasoning models (low C_0, zero gain) at a clean 20% threshold. This transforms SICA from "yet another ensemble method" into a method with formally characterizable applicability boundaries.

Limitations. Our evaluation focuses on logic reasoning (FOLIO, ProofWriter) where FOL constraint extraction is natural. Extending to domains where logical formalization is less straightforward (commonsense reasoning, open-ended generation) remains future work. The GSM8K pilot suggests C_0 transfers across domains, but constraint extraction quality may vary. Statistical significance on individual 60-question subsets is limited; the full-scale 804-question evaluation provides the primary evidence.

References

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A GSM8K Pilot Details

Table 6: GSM8K pilot results (Qwen2.5-14B, $K = 12$, 30 questions).

	SC	SICA	C_0
GSM8K	90.0	90.0	10%

Only 3 of 30 problems produce multiple candidate answers. Constraint extraction rate is 88%—slightly lower than logic benchmarks (95%) because arithmetic reasoning chains contain more implicit steps. The null result confirms that C_0 is task-relative: the same 14B model shows high C_0 on formal logic (where it is uncertain) and low C_0 on arithmetic (where it is confident).

B Full Constraint Ablation

On the 22 problems where MAX-SAT and count-only disagree, MAX-SAT wins 14 (63.6%), count-only wins 5 (22.7%), and 3 are ties (13.6%). MAX-SAT's advantage comes from detecting internal contradictions in constraint sets supporting incorrect answers—contradictions that constraint counting ignores.

C QwQ-32B Detailed Results

Table 7: QwQ-32B per-benchmark breakdown (48 questions).

Benchmark	n	C_0	Gap
AIME	22	0.0%	0.0
FOLIO	10	0.0%	0.0
ProofWriter	16	18.75%	0.0
Combined	48	3.6%	0.0

The 96.4% unanimous voting rate across problems confirms that QwQ-32B's reasoning chain self-correction produces convergent outputs regardless of sampling temperature.